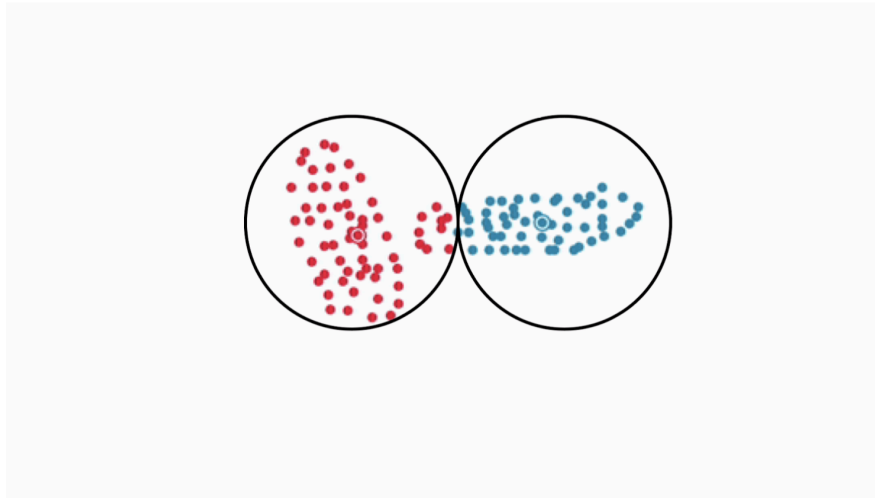


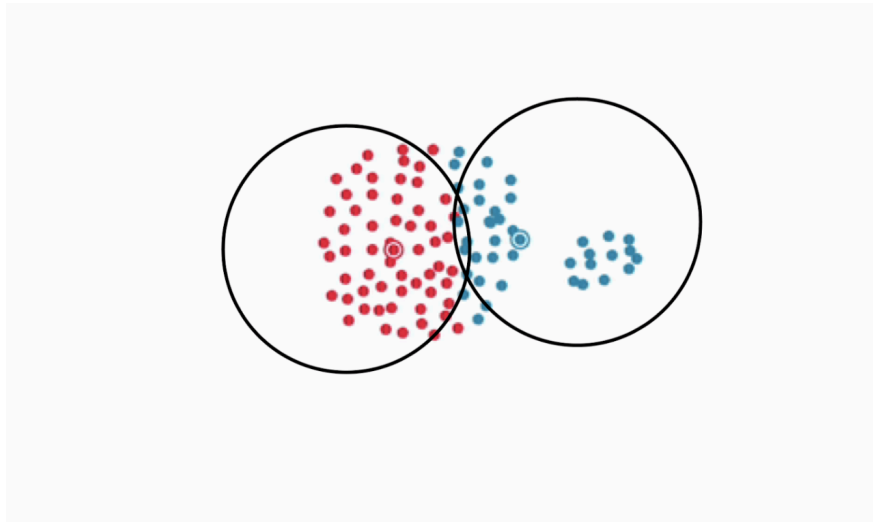
GMM

Problems with Kmeans Clustering

1. It creates a Globular Structure



2. Doesn't work well with clusters of different densities



3. Hard Clustering

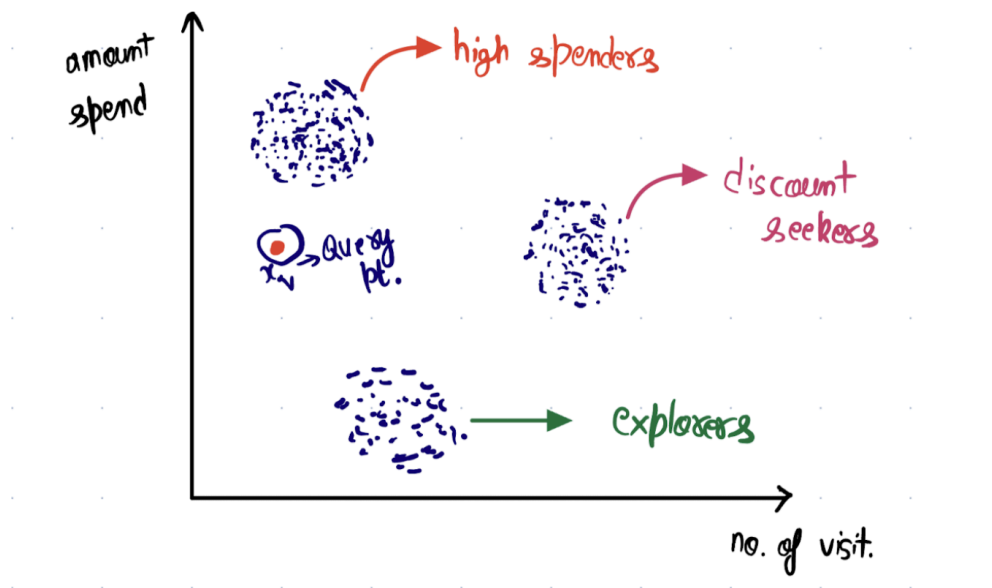
Datapoint is assigned to a single cluster only.

Why is hard clustering an issue?

It is possible that a data point can exhibit properties of multiple clusters.

For example:

A query point



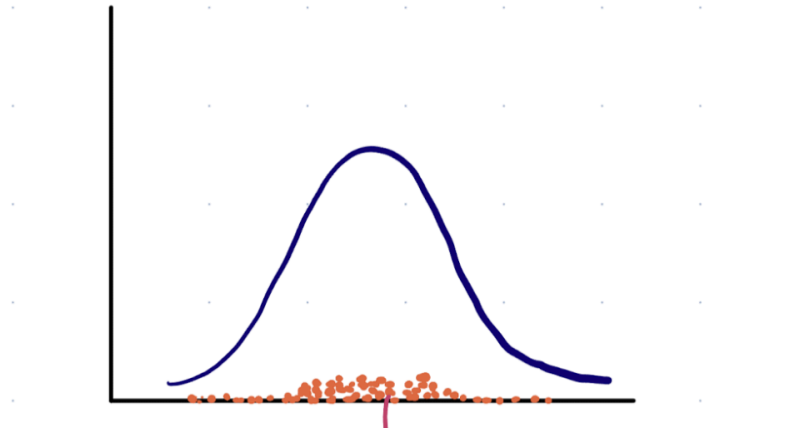
A customer can be both wealthy and still be price-conscious. Let's say,

- he/she can fall in the "high spender" cluster with a probability of 60%,
- but in the "price-conscious" cluster with a probability of 40%.

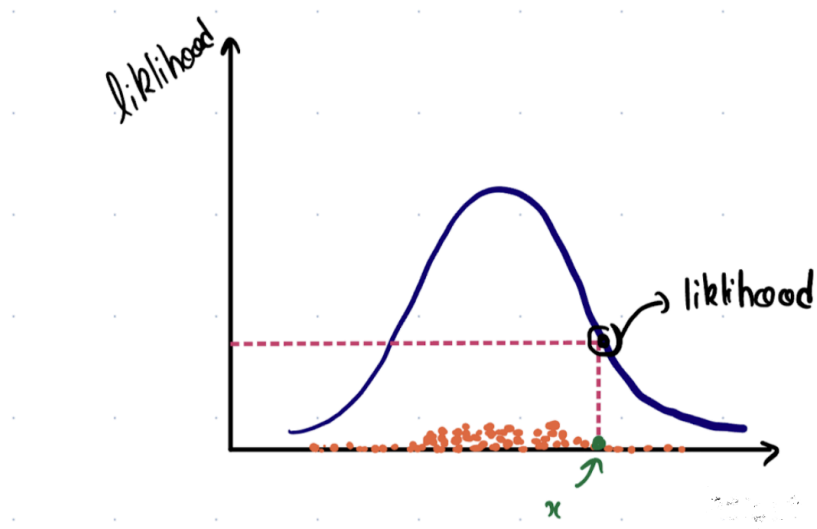
i.e. There is a probability attached to datapoint belonging to a cluster.

This concept is called **"Soft clustering"**

1-D Gaussian



- Mean (μ) tells us the position of the gaussian distribution
- where as, std dev (σ) tells us the shape.
- The likelihood of datapoint x is given as the
 - y value corresponding to that x i.e. y component of that instance.



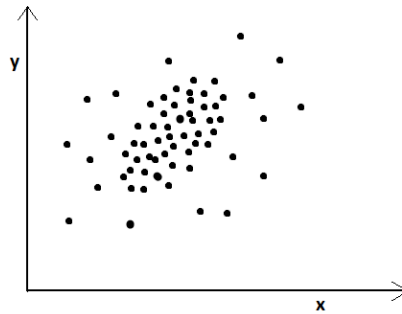
Mathematically, likelihood is calculated as:

$$\underbrace{p(x; \mu, \sigma)}_{\text{PDF}} = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

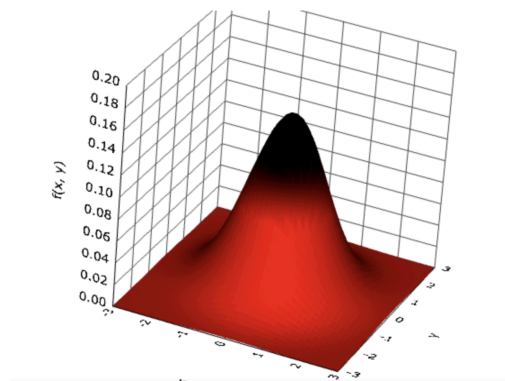
2D Gaussian

Viz: [link](#)

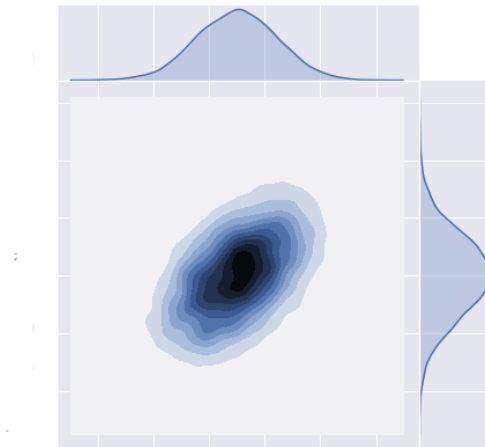
Let's consider two variables x and y with the following 2D scatter plot.



The Gaussian distribution for the following will be as follows:



If we were to look at it from the top, we would get its contour plot as:



- The part where the color is dark has more density (probability) of points and vice versa.
- As the distribution spreads outwards,
 - datapoints on the inner ring will have a higher likelihood as compared to datapoints belonging to the outer ring.
 - However, all the datapoints on a ring will have the same likelihood.
- The above multivariate Gaussian distribution is defined by five parameters:
 1. μ_x - the mean of the x-coordinate.
 2. μ_y - mean of y-coordinate
 3. σ_x^2 - variance in x direction.
 4. σ_y^2 - variance in the y direction.
 5. **Cov(x,y)** - Covariance between x and y.

We define the likelihood function as:

PDF:

$$p(x) = \frac{1}{|\sqrt{2\pi\Sigma}|} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

$\vec{\mu} \rightarrow$ mean vector $\rightarrow \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$ $\downarrow$ 2x1

we have 2 variables (x & y)
one mean for each variable

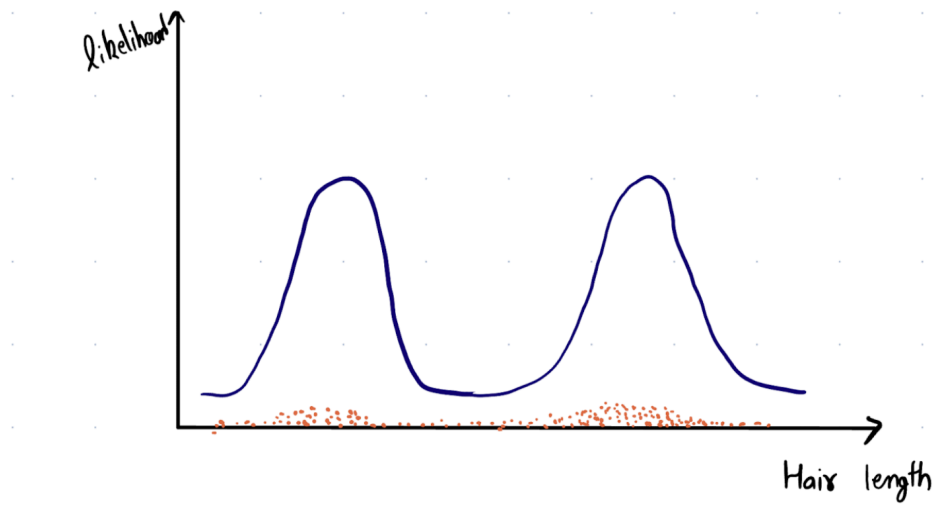
$\Sigma \rightarrow$ covariance matrix $\rightarrow \begin{bmatrix} \sigma_x^2 & \text{cov}(x,y) \\ \text{cov}(x,y) & \sigma_y^2 \end{bmatrix}$ $\downarrow$ 2dim 2×2

Visualization: [link](#)

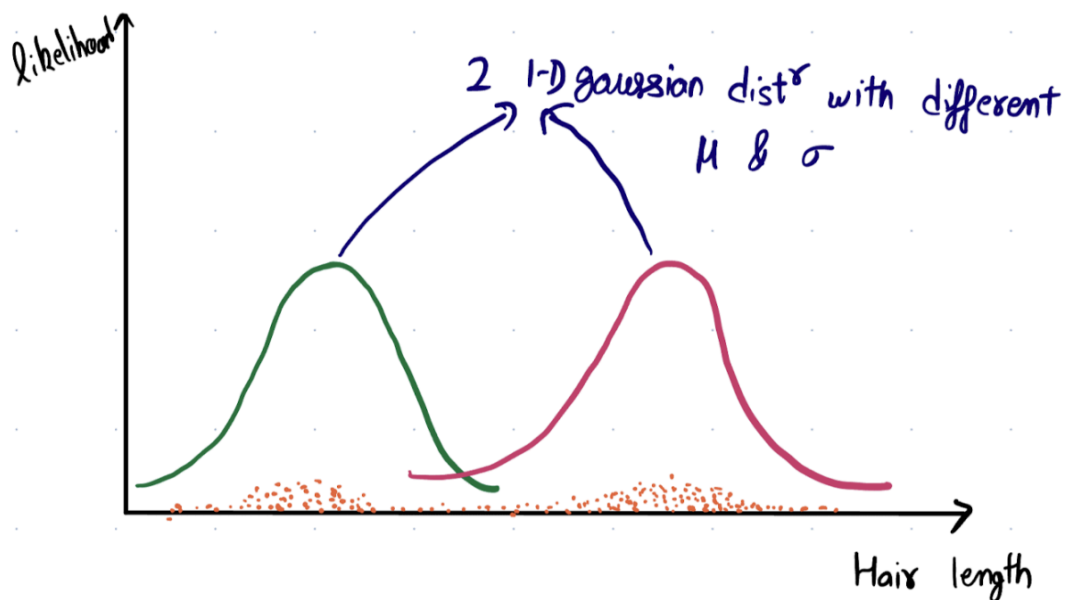
Gaussian Mixture Model

Assume we are trying to evaluate the effectiveness of a new topical treatment for hair loss

This trial involves recording the hair length of various participants.



We can use two 1D Gaussian distributions to model this data.



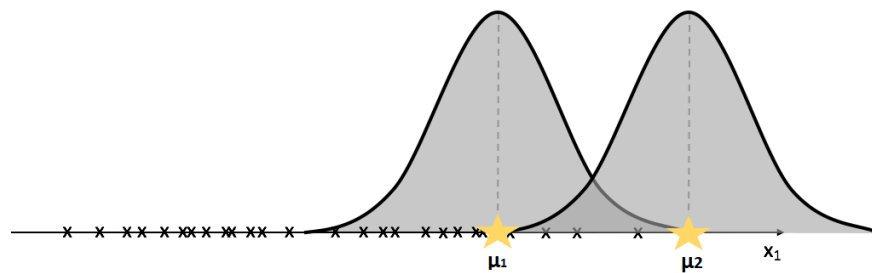
The idea behind GMM is:

- We try to fit the Gaussian distribution to each of the data clusters and check
- if datapoint belongs to the Gaussian distribution
 - it means it belongs to the cluster as well.

i.e. More the likelihood of a datapoint belonging to the distribution, the more we can say that the datapoint belongs to the cluster.

Expectation Maximization Algo

→ Randomly initialize μ and σ^2 .



→ Update μ and σ^2 .

- Calculate the probability of each point belonging to each Gaussian

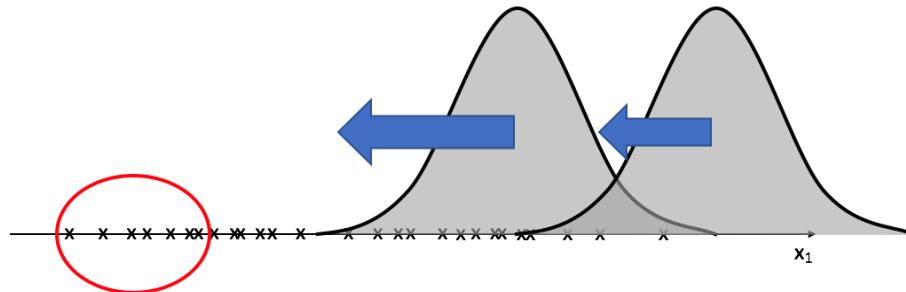
$$N(x | \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x - \mu)^2}{2\sigma^2}}$$

using the formula:

- Assign all the points which have the highest probability to their respective Gaussians.
- Now, compute new μ for each Gaussian as the mean of the points belonging to that Gaussian
- Similarly, update the variance.
- Here μ is updated as the **weighted average** of the points belonging to a Gaussian against a simple average which we followed in K-Means.

$\mu_j = 1/n \sum x_i \cdot (P_i(x_i))$, where n is the number of points with a non-zero probability of belonging to a cluster and $P_i(x_i)$ represents the probability of i^{th} point belonging to the j^{th} cluster.

- σ^2 of the i^{th} cluster is similarly updated in a weighted manner.



→ After multiple updates, we will have tightly fitting Gaussian distributions representing different clusters.

Visualization: [link](#), [link](#)

Kmeans vs GMM

k-Means	GMM
Clusters are defined by their means	Clusters are defined by their means and their variance, modeled as Gaussians
Limitations if clusters are overlapping	Works if clusters are overlapping
Uses Euclidean distance to the mean	Uses the probability of X belonging to a cluster (generative)

- If all the variances are equal i.e $\sigma^2_1 = \sigma^2_2 = \sigma^2_3$ and covariance, $\rho=0$ then,
 - we'll get circular clusters just like we get in Kmeans